Incidental Requirements - Incident Escalation & Severity Management (Viet Tran)

1. Scope:

This document captures functional and non-functional requirements for the **Incident Escalation and Severity Management** component of *Incidental*, a real-time incident reporting and management system.

2. Fully Dressed Use Cases (Escalation & Severity Management)

Use Case UC-ESM-01: Assign Responder to Incident

Primary Actor: Incident Commander

Supporting Actors: System

Stakeholders and Interests

- Incident Commander: Wants to quickly assign engineers or teams to an active incident.
- Responders: Want to be notified promptly when assigned so they can begin work.
- Managers: Want accurate records of who is responsible for an incident at any given time.

Preconditions

- An incident exists and is active.
- User is authenticated and authorized to assign responders.

Success Guarantee (Postconditions)

- Selected responders are linked to the incident record.
- Each assigned responder receives a notification.
- The assignment is logged in assignment history with a timestamp.

Main Success Scenario

1. Actor navigates to the active incident.
2. Actor selects "Assign Responder."
3. System displays available engineers, teams, and on-call groups.
4. Actor selects one or more responders.
5. System validates and assigns the responders to the incident.
6. System notifies assigned responders and updates the incident view for all active users.
7. System logs the action to assignment history.

Extensions

- 4a. Actor selects an on-call group:
    1. System resolves the group to the currently active on-call engineer.
    2. Resume from step 5.
- 5a. Responder is already assigned:
    1. System alerts the actor and skips the duplicate.

Special Requirements

- Assignment updates must propagate to all active users within 2 seconds.

Technology and Data Variations

- Responders may be individuals, teams, or on-call groups from a scheduling integration.

Frequency of Occurrence: High during active incidents.

Use Case UC-ESM-02: Reassign or Remove Responder

Primary Actor: Incident Commander

Supporting Actors: System

Stakeholders and Interests

- Incident Commander: Wants to update the roster during an incident due to shift changes or escalations.
- Managers/Auditors: Want all roster changes logged for accountability.

Preconditions

- An incident exists and is active.
- At least one responder is currently assigned.
- User is authorized to manage responders.

Success Guarantee (Postconditions)

- The incident's responder list reflects the changes made.
- Affected responders are notified.
- The change is logged in assignment history with a timestamp.

Main Success Scenario

1. Actor opens the active incident and navigates to the Responders panel.
2. Actor selects a responder and chooses "Reassign" or "Remove."
3. (For Reassign) System displays available responders; actor selects a replacement.

4. System updates the responder list and logs the change.
5. System notifies affected responders and updates the view for all active users.

Extensions

- 2a. Actor removes a responder with no replacement:
    1. System removes the responder and warns the actor if the incident now has zero assigned responders.
- 3a. Selected replacement is unavailable:
    1. System rejects the change and retains the original responder.

Special Requirements

- Roster updates must propagate to all active users within 2 seconds.

Technology and Data Variations

- Notifications may be delivered via email, Slack, or configured alerting integration.

Frequency of Occurrence: Medium/High during active incidents.

Use Case UC-ESM-03: View Current Responders

Primary Actor: User (Responder/Engineer/Stakeholder with access)

Supporting Actors: System

Stakeholders and Interests

- Responders: Want to know who else is working on the incident for coordination.
- Stakeholders: Want to know who is responsible for resolving the incident.

Preconditions

- An incident exists.
- User has permission to view the incident.

Success Guarantee (Postconditions)

- User sees the current list of assigned responders with names, roles, and assignment times.
- No system state is changed.

Main Success Scenario

1. User opens the incident detail view.
2. System retrieves and displays the current responder list.
3. Each entry shows the responder's name, role/team, and assignment timestamp.
4. The list refreshes automatically when assignment changes occur.

Extensions

- 2a. No responders are assigned:
    1. System displays a message indicating no responders are currently assigned.
- 2b. Data fails to load:
    1. System displays an error and offers a retry option.

Special Requirements

- The responder list must reflect changes within 2 seconds without a manual page reload.

Technology and Data Variations

- Sensitive contact details may be hidden for stakeholder roles per access control configuration.

Frequency of Occurrence: High during active incidents.

Use Case UC-ESM-04: View Assignment History

Primary Actor: Incident Commander, Auditor

Supporting Actors: System

Stakeholders and Interests

- Incident Commander: Wants a full record of assignment changes for situational awareness.
- Auditors/Managers: Want an immutable log for accountability and compliance.
- Postmortem Owners: Want complete history to support accurate incident reports.

Preconditions

- An incident exists (active or resolved).
- User has read access to the incident's history.

Success Guarantee (Postconditions)

- User sees a chronological log of all assignment, reassignment, and removal events.
- Each entry shows event type, responder name, acting user, and timestamp.

- No system state is changed.

Main Success Scenario

1. User opens the incident detail view and selects the "Assignment History" tab.
2. System retrieves and displays all assignment events in chronological order.
3. User can filter by date range or event type.
4. User optionally exports the history as a report.

Extensions

- 2a. No history exists:
    1. System displays a message stating no assignment history is available.
- 2b. Data fails to load:
    1. System displays an error and offers a retry option.

Special Requirements

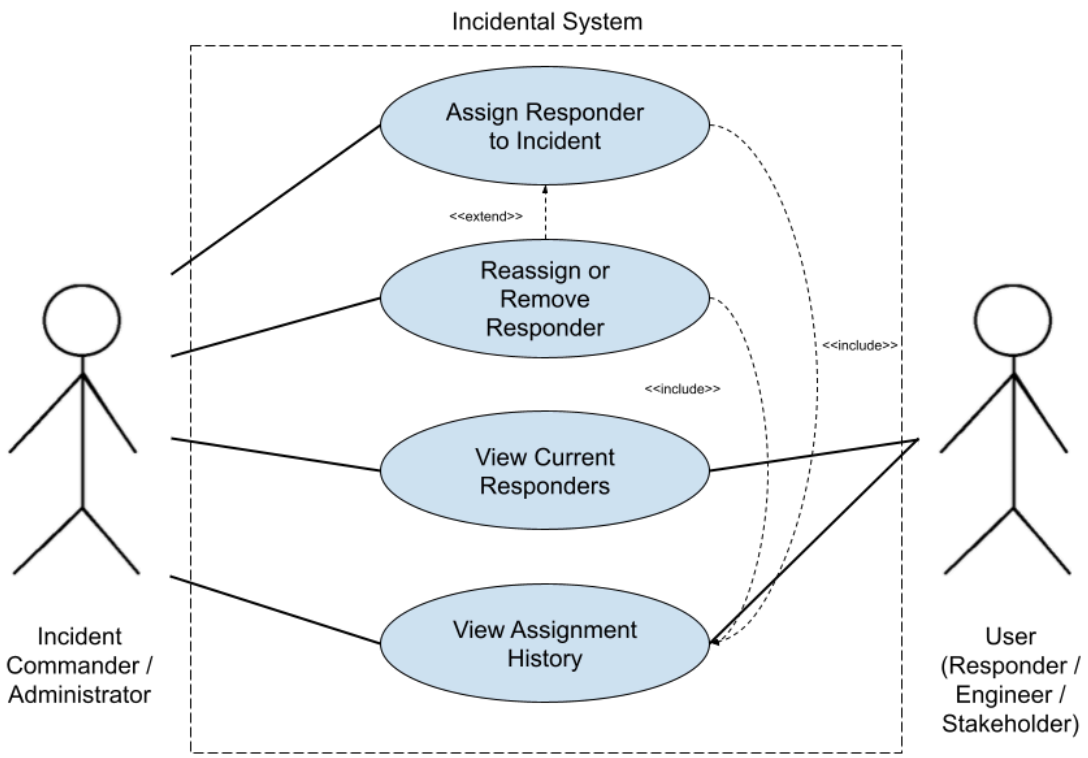
- Assignment history records must be immutable (append-only) once created.

Technology and Data Variations

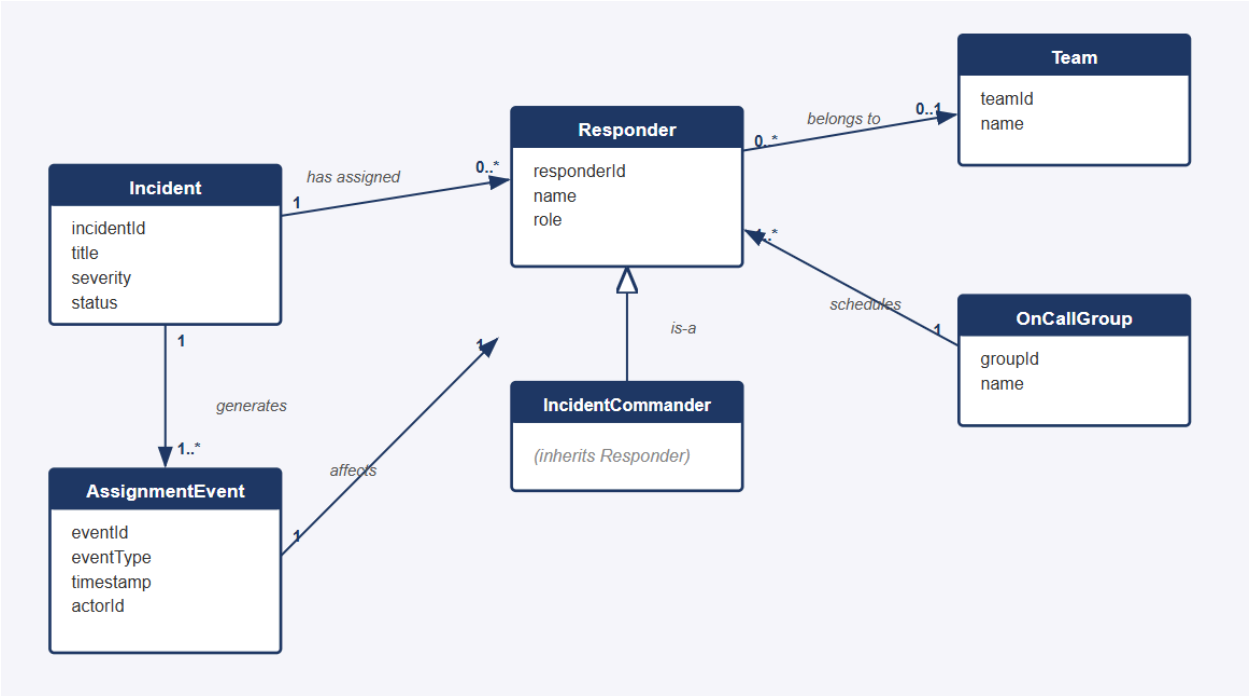
- History may be exported as CSV or PDF for postmortem or compliance use.

Frequency of Occurrence: Medium.

3. Use Case Diagram (Escalation & Severity Management)



4. Conceptual Class Diagram (Escalation & Severity Management Domain Model)



5. Supplementary Specifications (Non-Functional Requirements)

NFR-ESM-1 (Performance / Real-Time Propagation): Responder assignment updates must be reflected to all active users within 2 seconds under normal operating conditions.

Explanation: This applies to all assignment, reassignment, and removal events. Compliance is measured from when the change is accepted by the system to when it becomes visible to other active users. The system must use a real-time push mechanism such as WebSockets or Server-Sent Events to achieve this, ensuring all participants have a consistent view of the responder roster at all times.